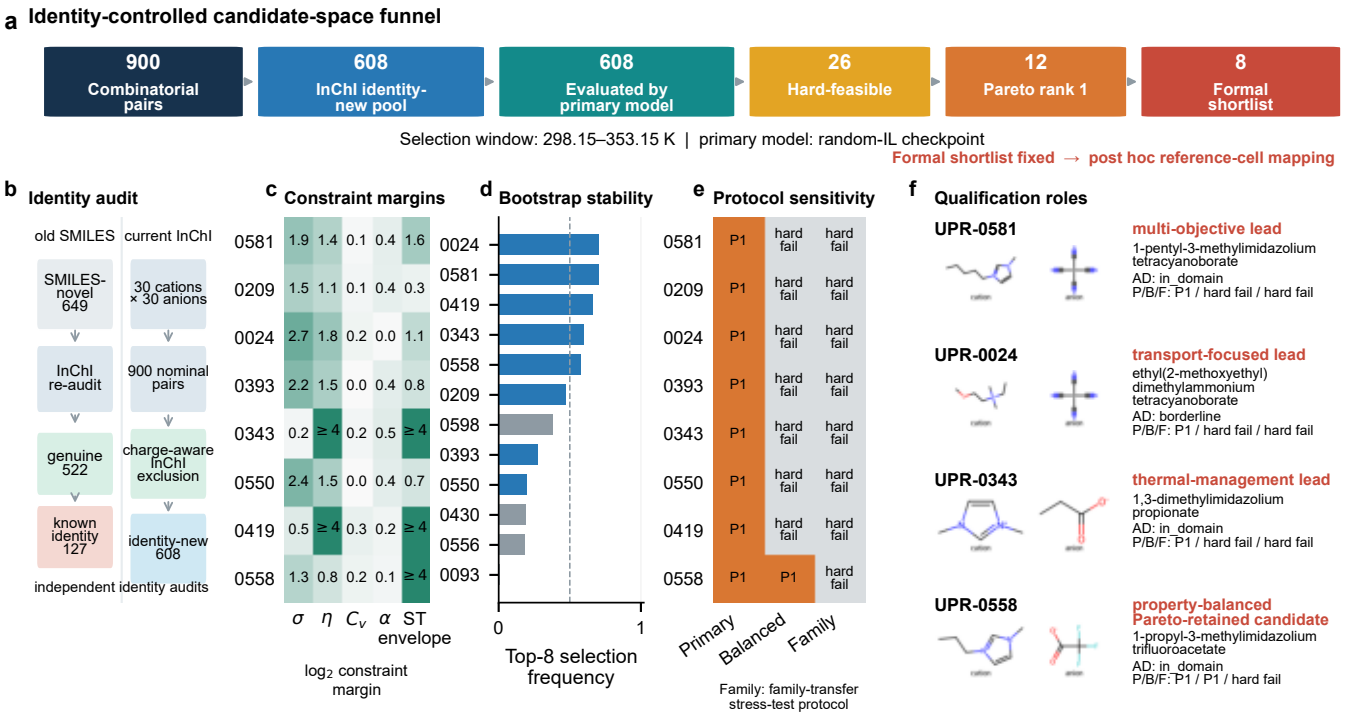

a
Identity-controlled candidate-space funnel
900
Combinatorial pairs
608
InChI identity-new pool
608
Evaluated by primary model
26
Hard-feasible
12
Pareto rank 1
8
Formal shortlist
Selection window: 298.15–353.15 K | primary model: random-IL checkpoint
Formal shortlist fixed → post hoc reference-cell mapping
b
Identity audit
old SMILES
current InChI
SMILES-novel 649
30 cations × 30 anions
InChI re-audit
900 nominal pairs
genuine 522
charge-aware InChI exclusion
known identity 127
identity-new 608
independent identity audits
c
Constraint margins
0581 1.9 1.4 0.1 0.4 1.6
0209 1.5 1.1 0.1 0.4 0.3
0024 2.7 1.8 0.2 0.0 1.1
0393 2.2 1.5 0.0 0.4 0.8
0343 0.2 ≥ 4 0.2 0.5 ≥ 4
0550 2.4 1.5 0.0 0.4 0.7
0419 0.5 ≥ 4 0.3 0.2 ≥ 4
0558 1.3 0.8 0.2 0.1 ≥ 4
σ η C_v α ST envelope
log2 constraint margin
d
Bootstrap stability
0024
0581
0419
0343
0558
0209
0598
0393
0550
0430
0556
0093
0
1
Top-8 selection frequency
e
Protocol sensitivity
0581 P1 hard fail hard fail
0209 P1 hard fail hard fail
0024 P1 hard fail hard fail
0393 P1 hard fail hard fail
0343 P1 hard fail hard fail
0550 P1 hard fail hard fail
0419 P1 hard fail hard fail
0558 P1 P1 hard fail
Primary
Balanced
Family
Family: family-transfer stress-test protocol
f
Qualification roles
UPR-0581
multi-objective lead
1-pentyl-3-methylimidazolium tetracyanoborate
AD: in_domain
P/B/F: P1 / hard fail / hard fail
UPR-0024
transport-focused lead
ethyl(2-methoxyethyl) dimethylammonium tetracyanoborate
AD: borderline
P/B/F: P1 / hard fail / hard fail
UPR-0343
thermal-management lead
1,3-dimethylimidazolium propionate
AD: in_domain
P/B/F: P1 / hard fail / hard fail
UPR-0558
property-balanced Pareto-retained candidate
1-propyl-3-methylimidazolium trifluoroacetate
AD: in_domain
P/B/F: P1 / P1 / hard fail